

FATIMA AZEEMI

Karachi, Pakistan azeemifatima1@gmail.com +923121118969 [GitHub](#) [LinkedIn](#) [medium](#)

SUMMARY

Computer Science graduate specialized in Artificial Intelligence with hands-on experience building AI-powered applications using Python, LLMs, RAG, NLP, and computer vision. Experienced in developing end-to-end AI solutions, integrating APIs and vector databases, and building full-stack applications using FastAPI, React, and Next.js.

WORK EXPERIENCE

AI Intern - Folio3 | Oct 2025 - Nov 2025

- Developed RAG pipelines using Python, LangChain, FAISS, and OpenAI embeddings for domain-specific product knowledge retrieval.
- Built an NLP-based tool to extract and rank high-performing product features from customer reviews using stemming, lemmatization, and text-processing techniques.

AI Fellow - Headstarter AI | Summer 2024

- Developed and integrated AI-powered applications using generative AI, computer vision, and text-to-speech models.
- Implemented RAG-based systems and integrated LLMs into web applications.
- Deployed AI applications using modern web frameworks and cloud platforms.

Frontend Intern - Metatalent.ai | Dec 2024 - Feb 2025

- Developed frontend components using React.js, Next.js, and CSS, collaborating in agile sprints and managing version control via GitLab.
- Documented and tested 100+ UI components with Storybook, performed manual testing, and validated user flows to support developer onboarding and project continuity.

SKILLS

- **AI & LLM:** LLMs, Generative AI, RAG, LangChain, Prompt Engineering, LLM Fine-Tuning, Vector Databases (FAISS, ChromaDB), NLP, Computer Vision
- **Programming:** Python, JavaScript, C++
- **Backend & APIs:** FastAPI, REST APIs, Node.js, Express
- **Frontend:** React.js, Next.js, Tailwind CSS, Material UI
- **Databases & Tools:** SQL, MongoDB, Git, GitLab
- **Cloud & Deployment:** Google Cloud Platform (GCP), Kubernetes/GKE, Cloud Run, VPC, IAM, Terraform, Cloud Monitoring - [Skill Badges](#)

EDUCATION

B.S. Computer Science (Specialization in Artificial Intelligence) | CGPA: **3.74 / 4.0**

NED University of Engineering and Technology | 2022–2026

PROJECTS

RAG - Scriptify | [GitHub](#)

- Built Scriptify, an offline privacy first, AI writing assistant for researchers using Python, CustomTkinter, Qwen, and ChromaDB with spell and grammar checking, LaTeX, and document Q&A.
- Implemented local RAG, PDF/DOCX processing, and background indexing to enable fast, private, offline AI assistance

Computer Vision - Fingerprint Quality Detector Web App | [GitHub](#) - [Website](#)

- Developed a web application using CNN (95% accuracy) and SVM models to classify fingerprint image quality.
- Trained models on augmented grayscale datasets and deployed using Hugging Face and Vercel.

Computer Vision - Inventory Website with Image Classification | [GitHub](#) - [Website](#)

- Developed a real-time inventory tracking system using computer vision model (90% accuracy) trained via Roboflow for object detection and classification.
- Integrated AI inference with a React-based interface and Firestore backend to automate inventory updates and enable intelligent querying.

RAG - YouTube QA Chatbot | [GitHub](#)

- Built a chatbot using LangChain, ChromaDB, and Groq AI to summarize long YouTube transcripts and answer user queries.
- Implemented an end-to-end RAG pipeline including transcript extraction, embedding generation, and semantic search.

Model Fine-Tuning using AWS Bedrock API | [GitHub](#)

- Fine-tuned a foundation model on domain-specific text data using AWS Bedrock and SageMaker APIs.
- Evaluated response quality and token efficiency, integrating the tuned model into a scalable inference pipeline.

AI Task Manager App | [GitHub](#)

- Developed a full-stack productivity app using the MERN stack with JWT authentication.
- Implemented AI-powered To-Do suggestions via Groq AI, task dashboard, and automated email reminders.

CERTIFICATIONS

- **Coursera:** Deep Learning Specialization, Mathematics for ML, TensorFlow Developer Certificate
- **Datacamp:** Associate AI Engineer for Developers Track
- **Udacity:** Introducing Generative AI with AWS
- **IEC:** Data Analytics

ACHIEVEMENTS

- **Udacity AI Programming with Python Nanodegree** - Scholarship recipient
- **AWS DeepRacer 2024** - Implemented a reinforcement learning algorithm for autonomous racing simulation; ranked among top 10 participants in Pakistan.
- **NSCT** - ranked in the **99th percentile nationwide** in the National Skill Competency Test conducted by HEC.